

SLIP – 3

Section I: Data Structures [15 Marks]

Q 1) Accept the array of n integers from user and sort the array in ascending order by using recursive Quick sort algorithm.

Answer:

```
#include <stdio.h>

void quicksort(int a[], int low, int high)
{
    int i = low, j = high, pivot = a[(low + high) / 2], temp;

    while(i <= j)
    {
        while(a[i] < pivot)
            i++;
        while(a[j] > pivot)
            j--;

        if(i <= j)
        {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
            i++;
            j--;
        }
    }

    if(low < j)
        quicksort(a, low, j);
    if(i < high)
        quicksort(a, i, high);
}

int main()
{
    int a[100], n, i;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
    }
    return 0;
}
```

```

    }

    printf("Enter %d integers:\n", n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quicksort(a, 0, n - 1);

    printf("Sorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (primary keys are underlined).

Person (pnumber, pname, birthdate, income)

Area (aname, area_type, pincode)

The relationship is as follows: Person-Area: many-to-one. The ‘area_type’ can have values as either ‘urban’ or ‘rural’.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Area(
    aname VARCHAR(50) PRIMARY KEY,
    area_type VARCHAR(10)
        CHECK(area_type IN ('urban','rural')),
    pincode INT
);

```

```

CREATE TABLE Person(
    pnumber INT PRIMARY
    KEY,
    pname VARCHAR(50),
    birthdate DATE,
    income NUMERIC(10,2),
    aname VARCHAR(50)
    REFERENCES Area(aname)
);

```

```

INSERT INTO Area VALUES
('Camp','urban',411001),

```

```
('kalyaninagar','urban',411006),  
('Wadgaon','rural',412201);
```

```
INSERT INTO Person VALUES  
(1,'Ravi','1990-07-15',30000,'Camp'),  
(2,'Sita','1992-03-10',50000,'kalyaninagar'),  
(3,'Ram','1988-07-21',20000,'Wadgaon');
```

Q 2(B)

B. Execute the following queries in PostGreSQL(any 3) [6 marks]

i. List the details of all people whose name starts with the alphabet 'R'.

Answer:

```
SELECT * FROM Person  
WHERE pname LIKE 'R%';
```

ii. List the names of all people whose birthday falls in the month of 'July'.

Answer:

```
SELECT pname FROM Person  
WHERE EXTRACT(MONTH FROM birthdate) = 7;
```

iii. Display the details of people in the sorted order of their income.

Answer:

```
SELECT * FROM Person ORDER  
BY income;
```

iv. Display the count of areas of 'urban' type.

Q 2(C)

C. Write a query to list the names of people who live in 'Camp' area and have income less than at least one person who lives in 'kalyaninagar' area. [4 marks]

Answer:

```
SELECT pname  
FROM Person  
WHERE aname = 'Camp'  
AND income < ANY  
(
```

```
SELECT income FROM Person
WHERE aname = 'kalyaninagar'
);
```

Q 3) Viva [5 marks]